Does Format Matter? Evaluating LLM Confidence Across Visualization Modalities

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Abstract

The integration of Large Language Models (LLMs) in data workflows has been extensively explored due to its potential to improve analytical tasks. Multimodal LLMs are increasingly used to interpret charts and visual data, but their reliability across input formats remains underexplored. This paper presents a controlled empirical study that compares three input modalities - rendered images, data tables, and descriptive paragraphs - across four types of continuous visualization and three LLMs. Our findings demonstrate that rendered image inputs consistently produced the worst confidence calibration, the lowest accuracy, and weaker hallucination resistance, with the strongest effects occurring in dense spatial encodings. This study also offers the Brier-ECR framework as a reusable protocol for assessing multimodal reliability and highlights the architectural bottlenecks that currently limit the effectiveness of visual formats in LLM-integrated analytic systems.

Index Terms: Large Language Models, information visualization, confidence calibration, hallucination resistance.

1 Introduction

A foundational premise of information visualization is that visual representation improves analytical performance in tasks such as trend detection and outlier identification for humans. Decades of research have supported the idea that well-designed visual encodings reduce cognitive load, enabling humans to recognize patterns and draw accurate conclusions more readily than from a data table or descriptive text alone [3, 4, 14]. Format is not a neutral container for information but an active contributor to interpretation [8].

Since the emergence of publicly accessible generative AI in 2022, LLMs have evolved into practical reasoning tools that users routinely consult for analytical tasks. Users can now submit data in a wide variety of forms—an image of a chart, a pasted data table, or a written description of a dataset—asking the model to identify patterns, detect anomalies, and draw conclusions. This flexibility has positioned LLMs as decision-support tools in contexts that often required statistical expertise, yet the choice of input format has received relatively little empirical attention.

This gap is significant because LLMs process continuous data through an architectural pipeline fundamentally different from human perception. While human chart comprehension relies on preattentive spatial processing [14] to recognize patterns, multimodal LLMs convert pixel arrays into token embeddings through patch-based vision transformers rather than sub-patch-level spatial structure [5]. Prior work has shown that LLMs struggle with spatially complex charts [13, 15], but whether this produces systematic differences in confidence calibration across modalities, and whether those differences interact with visualization type, has not been directly examined. We address this gap with a controlled study evaluating three input modalities across four continuous visualization types, finding that rendered image inputs consistently produced the weakest confidence calibration and reduced hallucination resistance, especially under dense spatial encodings.

2 Related Work

Human Perception in Visual Encodings. Prior visualization research has established that representational format shapes analytical performance. Cleveland and McGill [3, 4] demonstrated that visual encodings such as position and length supported more accurate perception than area and color encodings. Ware [14] further argued that well-designed encodings reduce cognitive load for pattern recognition. Together, these findings suggest that the format actively shapes analytical performance for humans. Whether LLMs follow a similar pattern is less understood.

LLM Perception in Visual Encodings. Recent work increasingly examined multimodal chart understanding alongside text-based reasoning. Masry et al. [9] introduced the ChartQA benchmark, showing that question-answering systems still struggle with complex data queries compared to human baselines. However, this literature predominantly focuses on discrete chart types such as bar and pie charts. Continuous data representations, such as the spatial encoding types identified by Tory and Möller [12], remain underexplored in LLM evaluation. Xu and Wall [15] found that input format affects LLM performance on low-level visual analytic tasks, with SVG-based representations mitigating some perceptual errors. Our work builds on this by introducing a three-way modality comparison specifically for continuous data.

Calibration and Hallucination in Visual Reasoning. The studies above suggest that visual formats can benefit human interpretation while LLMs may struggle with the same inputs. What remains unclear is how input modality affects reliability. Accuracy alone is insufficient for analytic use: an incorrect answer is more consequential when expressed with high confidence. Prior work has shown that LLMs exhibit systematic miscalibration that depends heavily on prompt format and context [16], but whether the relationship between confidence and accuracy extends to visual modalities remains uncertain. This study therefore uses Brier score [2] to measure calibration, as it penalizes confident errors more than uncertain errors. A related concern is grounding. Ji et al. [6] defined hallucination as output not grounded in the available evidence from the input. To test this failure mode, we included Explicit Claim Rejection (ECR) tasks, which present models with false claims and score their confidence-calibrated rejection with the same Brier score framework. Together, these analyses allow us to evaluate whether input modality affects both confidence calibration on analytical tasks and hallucination resistance.

3 Method

In this study, we evaluate how representational format influences LLM analytical performance and confidence calibration across continuous visualization tasks. We designed an experiment that accesses 3 input modalities $\{$rendered image, data table, descriptive paragraph$\}$ $\times$ 4 visualization types $\{$Line Graph, Color Map, Isoline, Arrow Glyph$\}$ $\times$ 3 LLMs $\{$GPT-5.4, Claude 4.6 Sonnet, Gem-

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ini 3.1 Flash}. For each of the 40 datasets, models performed three analytical tasks—value retrieval, trend detection, and outlier detection—and three Explicit Claim Rejection (ECR) tasks to measure hallucination resistance. We provided the formatted data and standardized prompts as input and recorded both the binary accuracy and numerical confidence ratings. We detail the full methodology in this section and report on our findings in the next section.

Visualization Type Selection. Selection is grounded in the model-based taxonomy of Tory and Möller [12], which classifies techniques according to their underlying data model. This study focuses on the continuous category, where data varies smoothly across spatial dimensions. These formats demand more precise interpolation and spatial reasoning than discrete charts.

From the nine subtypes identified by Tory and Möller [12], we excluded five types—direct volume rendering, isosurfaces, particle traces, line integral convolution, and tensor ellipsoids—because their semantics do not translate cleanly across modalities. These types rely on visual conventions such as opacity, flow, and convolution texture that cannot be reduced to structured tables or text without losing essential spatial depth. Including them would introduce interpretive bias and reduce comparability.

The four remaining types span a meaningful range of spatial encodings encountered in general analytical work.

- **Line Graphs:** Encode one variable along a spatial axis.
- **Color Maps:** Encode magnitude via chromatic gradients.
- **Isolines:** Represent scalar fields requiring boundary identification.
- **Arrow Glyphs:** Encode vector fields via simultaneous direction and magnitude, imposing the highest perceptual load.

Datasets and Stimuli. For each visualization type, 10 distinct datasets were selected from publicly available online sources via a randomized process, yielding 40 dataset-type pairings. Datasets varied in domain and structural properties to support replication across conditions within each visualization type. All datasets are explicitly labeled fictional in the prompt to prevent LLMs from drawing on memorized real-world associations.

After dataset construction, each data set was converted into three different input modalities: rendered image, data table, and descriptive paragraph. A standardized Python script, available in the supplemental materials, was used to convert data tables (in CSV) into descriptive paragraphs (in TXT) and rendered images (in PNG), all available in supplemental materials. The underlying numerical values, labels, units, and dataset structure were preserved so the difference in model performance can be attributed to input modality.

- **Data table (DAT):** The numerical values were presented as a structured data table with labeled rows and columns in a CSV file containing a metadata header indicating the title, row/column label, entity, year, and units.
- **Rendered image (VIS):** Each data set was presented as a rendered PNG image generated from the CSV file at 300 DPI with standardized dimensions. Figures include title, axis labels, legends, units, and tick labels. Color palettes and marker encodings were applied consistently across figures to support visual comparison.
- **Descriptive paragraph (PAR):** Each data table was converted into a natural-language description that represented the same numerical content and structural relationships shown in the rendered image and data table.

Tasks. Each of the three input modalities for the 40 datasets was paired with the same six standardized questions: three scored analytical tasks and three Explicit Claim Rejection (ECR) items. Analytical tasks were grounded in the low-level task taxonomy of Amar et al. [1]. Q1 targeted value retrieval—identifying a specific numerical value at a given location or condition. Q2 required trend detection—identifying a directional or relational pattern across the dataset. Q3 assessed outlier detection—identifying values or regions that deviate from the dominant pattern. For Arrow Glyph datasets, Q1 was adapted to assess directional inference rather than precise numerical retrieval, given that vector field readings require integrating magnitude and direction simultaneously. Each ECR item presented the model with a specific false claim—an incorrect maximum value (ECR1), a fabricated trend (ECR2), or a non-existent outlier (ECR3)—and asked the model to accept or reject it. ECR items were constructed by taking the correct answer to each analytical question and substituting a plausible but incorrect alternative. All questions were administered using a standardized prompt (see supplemental materials) instructing the model to provide both a response and a numerical confidence rating from 0 to 100, rescaled to 0–1 for scoring.

Procedure and Data Collection. Prompts were administered via API to GPT-5.4, Claude 4.6 Sonnet, and Gemini 3.1 Flash at a temperature of 1.0, with exact consistency across conditions. Response logs—including each model's stated answer and confidence rating per sub-task were recorded. All scripts and raw response logs are included in the supplemental materials. The fully crossed design—3 modalities $\times$ 4 visualization types $\times$ 10 datasets $\times$ 3 LLMs—yielded 360 scored trials, each comprising three analytical sub-tasks and three ECR items and their corresponding confidence score.

Metrics. Three outcome measures were computed for each trial condition. *Accuracy* was computed as the binary mean across the three analytical sub-tasks, scored against a pre-defined ground truth answer key established before data collection. *Confidence calibration* was measured using the Brier score [2]:

$$BS = (confidence - outcome)^2$$

where confidence is the model's stated probability (0–1) and outcome is the binary correctness indicator (1 = correct, 0 = incorrect), averaged across sub-tasks. Lower Brier scores indicate better calibration; for binary outcomes, 0.25 corresponds to assigning 0.5 confidence, the maximally uncertain prediction. *Hallucination resistance* was measured using the same Brier score framework on ECR tasks. For each planted false claim, correct rejection was scored as 1 while acceptance, failure to reject, or ambiguous rejection was scored as 0. The resulting ECR Brier score captures whether models reject unsupported claims with appropriately calibrated confidence. The paired analytical task and ECR task Brier approach was designed to detect cases where models are both wrong and overconfident, which accuracy alone would obscure.

Inference Model Selection. Each metric was analyzed using inferential models (full results in supplemental materials), with model structure determined by the level of analysis. Dataset identity was included as a random intercept in the main modality and VIS-specific ECR models to isolate condition effects from baseline difficulty. Task-level models omit this term because task type is fully crossed with vis-type and LLM at that level, spreading dataset variance evenly across conditions. Models therefore account for residual variance differently and should not be directly compared.

4 Results

In this section, we present the results of the LLMs' performance and calibration across modalities using binary accuracy and the Brier score. We structure this section by first discussing the overarching effect of input modality, followed by the impact of visualization type and specific task categories.

Input Modality Effect on Accuracy and Calibration. We fit linear mixed-effect models (LMEMs) predicting task-averaged Brier score and accuracy from input modality (modality), visualization type (vis-type), and LLM model (LLMs). One entry (dataID: PAR-2I) was excluded due to a Claude API refusal to respond, leaving

Group Label	EST DIFF (Brier)	P (Brier)	EST DIFF (Acc)	P (Acc)
Baseline DAT + Glyph + Claude	0 (ref.)	—	0 (ref.)	—
PAR	-0.004	.702	**-0.054**	**.008**
VIS	**0.039**	**< .001**	**-0.044**	**.028**
Color Map	-0.013	.449	**-0.067**	**.005**
Isoline	-0.030	.087	0.022	.351
Line Graph	**-0.055**	**.002**	**0.061**	**.010**
GPT	-0.016	.081	**-0.045**	**.028**
Gemini	**0.025**	**.007**	-0.029	.150

Figure 1: Linear mixed-effects model for mean Brier score and task accuracy from input modality, visualization type, and LLM model. Estimations are reported as differences relative to the baseline condition (DAT+Arrow Glyphs+Claude). Positive Brier estimates indicate worse confidence calibration while positive accuracy estimates indicate higher task accuracy.

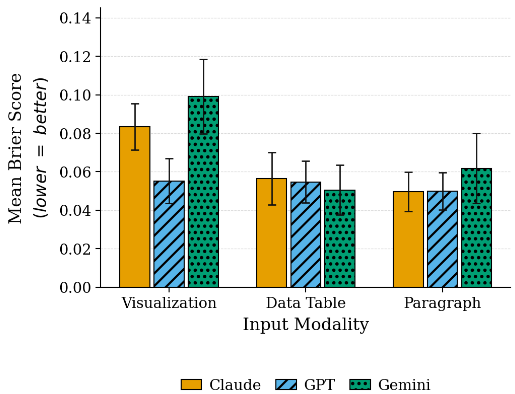

Figure 2: Mean Brier score by input modality and LLM. Error bars indicate ± 1 standard error of the mean (SEM). All experimental results fell below the 0.25 threshold that corresponds to maximally uncertain predictions.

359 observations. The baseline condition in both models was DAT+ Arrow Glyphs + Claude.

The Brier score model showed a significant modality effect. Relative to DAT inputs, VIS inputs produced significantly higher Brier scores (estimated difference from baseline, $\beta = +0.039, p < 0.001$), indicating a worse confidence calibration. In contrast, PAR inputs did not differ significantly from DAT inputs ($\beta = -0.004, p = 0.702$). The accuracy model showed a similar pattern. Relative to DAT, both PAR ($\beta = -0.054, p = 0.008$) and VIS ($\beta = -0.044, p = 0.028$) performed significantly worse. Results are summarized in Figures 1 and 2.

Visualization Type Effect on Accuracy and Calibration. Vis-type also affected model performance beyond modality alone. Descriptively, Arrow Glyphs consistently produced the highest Brier scores across all modalities while Line Graphs maintained the lowest (Figure 3). This trend is further supported by the LMEMs. Relative to baseline (Arrow Glyphs), Line Graphs showed significantly better Brier scores ($\beta = -0.055, p = 0.002$) and accuracy ($\beta = +0.061, p = 0.01$), while Color Maps performed significantly worse in accuracy ($\beta = -0.067, p = 0.005$). Color Map and Isoline did not differ significantly from baseline condition in calibration. These results suggest that simpler one-dimensional spatial encodings yield better model calibration.

Model Effect. LLM Model also contributed to performance differences. Gemini produced significantly worse Brier scores than Claude ($\beta = 0.025, p = 0.007$) while GPT did not differ significantly from Claude in calibration. In the accuracy model, GPT produced significantly lower accuracy than Claude ($\beta = -0.045, p = 0.028$) while Gemini did not differ significantly from Claude.

Dataset-Level Variance. Dataset-level variance was significant

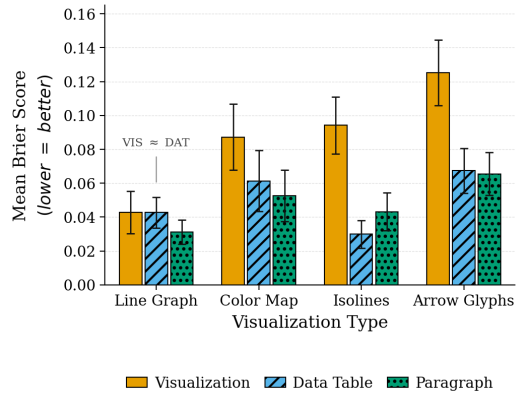

Figure 3: Mean Brier score by visualization type and input modality, across all three LLMs (error bars = ±1 SEM).

Question Type	Mean Brier	EST DIFF	P
Value Retrieval Baseline	0.073	0 (ref.)	—
Trend Detection	0.032	**-0.041**	**.024**
Outlier Detection	0.229	**0.156**	**< .001**
ECR	0.012	**-0.062**	**< .001**

Figure 4: Linear fixed-effects model result predict task Brier score from question type. Estimations are reported as differences from baseline condition (Value Retrieval).

for Brier score model ($var = 0.200, p = 0.007$), indicating that the broad analysis of modality, vis-type, and LLMs alone only partially explain the calibration performance. The accuracy model showed insignificant dataset-level variance ($var = 0.004, p = 0.889$), suggesting that accuracy differences can be mostly explained by these three conditions. The remaining variance in Brier score model motivated a follow-up analysis examining whether task type could account for the calibration performance.

Task Type Effect on Confidence Calibration. To examine this, we fit a linear fixed-effects model predicting question-level Brier score from task type, vis-type, and LLMs (see Figure 4). Relative to the baseline (value retrieval), trend detection and ECR items produced significantly lower Brier scores ($\beta = -0.041, p = 0.024$ and $\beta = -0.062, p < 0.001$, respectively) while outlier detection produced substantially worse Brier scores ($\beta = 0.156, p = 0.001$).

Task-specific Calibration Across Visualization Types. To determine if the task-level confidence calibration differences depended on vis-type, we fit an aggregate fixed-effects model predicting mean task-level Brier scores from task type (including average ECR), vis-type, their interaction, and LLMs. Outlier detection showed highest Brier scores across all vis-types and LLMs (Figure 5). The model also found a significant negative interaction between trend detection and Color Maps, indicating that this combination was better calibrated than expected from the separate effects from task type and vis-type ($\beta = -0.13, p = 0.003$). Overall, poor calibration was concentrated in tasks requiring localized spatial identification (value and anomaly identification) rather than global pattern recognition (trend detection).

Hallucination Resistance Across Task-Visualization Combinations. The ECR column in Figure 5 extends the task type effect to hallucination resistance. Average ECR Brier scores were generally lower than analytical tasks, suggesting LLMs are potentially more reliable when evaluating an explicit false claim compared to identifying values, trends, or outliers. Descriptively, Arrow Glyphs showed higher ECR Brier scores in several conditions. This finding further motivated analysis of ECR Brier scores restricted to VIS inputs.

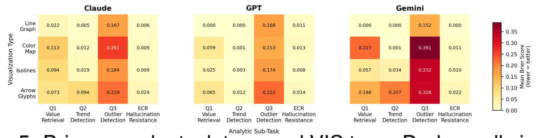

Figure 5: Brier score by task type and VIS type. Darker cells indicate higher/worse Brier scores (worse calibration). The ECR column reports hallucination-resistance performance using the average Brier score across False-Claim Rejection tasks. Inferences used an aggregate fixed-effect linear model with task type, VIS type, their interaction, and model as predictors.

Restricted to VIS inputs, a LMEM predicting ECR Brier score from vis-type, LLM, and ECR questions confirmed that Line Graphs produced significantly stronger hallucination resistance than Arrow Glyphs ($\beta = -0.010, p = 0.002$). Together, these results suggest that dense spatial encodings degrade both analytical confidence calibration and hallucination resistance, especially under VIS inputs. This finding is also supported descriptively with Isolines consistently showing high ECR Brier scores.

5 Discussion and Conclusion

Findings. The DAT advantage over VIS is consistent with the architectural differences between text and image processing in LLMs. Since tabular data can be represented as serialized text, it can be passed directly into the token-based pipeline used by LLMs [11], whereas rendered images require an additional step of recovering numerical data from spatial encodings before any reasoning. This extra step, commonly a vision transformer, divides image inputs into smaller patches where each patch is flattened and linearly projected into a single token embedding. An attention mechanism then operates over these tokens, leaving the models with no direct access to sub-patch information [5]. Furthermore, the standard attention mechanism computes relationships between each token pair equally regardless of spatial proximity in the original input. The combined effect leaves the models unable to reason precisely about local spatial context.

This architectural limitation is reflected in the task effect analysis. Outlier detection, which requires pinpointing a local anomalous region, produced the highest Brier scores across all conditions. In contrast, trend detection, which only requires recognition of a global pattern, remained comparatively well calibrated. The significant negative interaction between trend detection and Color Maps also suggests that chromatic encodings may partially offset the limitations, though the mechanism behind this requires further investigation.

Hallucination resistance results extends Kadavath et al.'s [7] findings that LLMs demonstrate a degree of self-knowledge in natural language tasks. ECR Brier scores were generally lower than analytical task scores, suggesting that evaluating an explicit false claim was an easier operation. This is consistent with Kadavath et al.'s [7] findings which suggests that models may be able to evaluate the false claims against their internal understanding of the input rather than conducting analytical tasks. However, this advantage was degraded for dense spatial encodings under VIS inputs, where Line Graphs produced significantly stronger hallucination resistance compared to Arrow Glyphs. A plausible explanation is the model's internal representation of input source for verification reference becomes less reliable as spatial complexity increases. In other words, the self-knowledge Kadavath et al [7] identified depends on the model having a reliable understanding as a reference for verification, a condition that dense visual encodings may undermine.

Limitations and Future Work. Four limitations bound the scope of our conclusions. First, the absence of a human baseline — precluded by ethics approval constraints — prevents direct comparison of LLM and human performance under the same conditions. Prior cognition research suggests humans benefit from visual encodings over raw tables [3, 4, 14]; whether the DAT advantage we observe represents a genuine inversion of human perceptual preferences or a property specific to token-based processing remains an open question.

Second, confidence scores were elicited linguistically rather than extracted as native log-probabilities. Because RLHF-trained models are incentivized to appear helpful and confident [10], verbalized scores may reflect communication tendencies rather than true internal uncertainty. Future work should validate the calibration signal by comparing verbalized scores against raw softmax probabilities from open-weight models.

Third, all conditions used a single standardized prompt across all three models. Results may be sensitive to prompt wording in ways our design cannot detect. Prior work [16] has shown that LLM outputs vary substantially across different prompts, and it is possible that a different prompt design would produce a different result. Future work should include prompt variation to determine how robust the DAT advantage is carried across different elicitation strategies.

Finally, while the consistency of results across GPT, Claude, and Gemini suggests the modality effect generalizes across LLMs, three models is an insufficient sample size to confirm universality. Future work should expand the LLM sample size, and include open-weight models to determine whether these effects are specific to RLHF-trained models, or reflect an architectural property of transformer-based visual processing.

Conclusion. This study found that rendered image inputs produced worse confidence calibration and lower accuracy than structured data tables and descriptive paragraphs across all models and visualization types. These results suggest that representational formats optimized for human perception do not necessarily transfer reliably to LLMs, a gap potentially rooted in token-based visual processing.

For developers, prompting rendered figures to LLMs risks producing analytical systems that are inaccurate and poorly calibrated. We offer the Brier-ECR framework as a reusable protocol for assessing multimodal reliability, capable of detecting miscalibration and hallucination susceptibility that accuracy alone would miss. All experimental materials are available in the supplemental materials to support further research.

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